

DATA ANALYSIS PROJECT REPORT

Project Title	Gaming Habits and Academic Performance Analysis
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Course / Section	Programming In Python (B)
Instructor	Tanzeem Rahat
Date of Submission	2 May 2026

1. Executive Summary

This project analyzes the relationship between gaming habits and academic performance using the Gaming vs Academic Performance dataset sourced from Kaggle, consisting of 8,000 student records. The main objective was to determine whether gaming harms academic outcomes directly or indirectly through related lifestyle factors such as sleep, study time, stress, and attendance. The analysis was conducted using Pandas for data processing, NumPy for normalization and custom metric creation, and Matplotlib for visual exploration.

A key contribution of the project was the construction of an Academic Risk Score, combining normalized gaming, study, and sleep hours. This score proved to be the strongest predictor of grades, showing a strong negative correlation with academic performance. While gaming was found to significantly improve reaction time, this cognitive benefit did not translate into higher grades. Additionally, gaming addiction showed no meaningful relationship with class attendance, indicating that poor performance is not caused by lack of absence in class. Overall, the findings suggest that academic outcomes are most strongly influenced by lifestyle balance, where excessive gaming combined with reduced sleep and study time leads to lower academic success, rather than gaming alone.

2. Introduction and Project Objectives

This study looks at how gaming affects academic performance and whether gaming lowers grades directly or by taking time away from study, sleep, and daily routines.

Item	Your Content
Dataset name	Gaming vs Academic Performance
Dataset source	https://www.kaggle.com/datasets/aiexplorer77/gaming-vs-academic-performance
Main project goal	To determine how gaming as well as study habits collectively influence academic performance
Research Question 1	Does an "Academic Risk Score" predict grades better than individual factors like study or sleep hours alone?
Research Question 2	Does improved reaction time from gaming translate into measurable benefits for academic performance?
Research Question 3	Does high gaming addiction correlate with a drop in school attendance?
Research Question 4	Does gaming reduce stress at the cost of grades, or does higher academic performance occur alongside increased stress?
Research Question 5	Does the type of gaming genre a student plays have a meaningful impact on academic performance?

3. Dataset Description

3.1 Basic Dataset Information

Property	Value
Number of rows	8000
Number of columns	14
Data types present	Numerical (float64, int64) and Categorical (object)
Target or key variable(s)	Grades
Potential issues observed at first glance	None observed (no null values), but outliers in grades were adjusted.

3.2 Initial Observations

The dataset is clean and well organized, with no missing values or duplicate records. It includes different types of information such as reaction time, gaming addiction score, and stress level. Because of this variety, the data allows us to analyze student behavior and performance from multiple angles, not just how much time students spend gaming.

4. Data Cleaning and Preparation

Standardized cleaning procedures were implemented to ensure the analysis is robust and accurate.

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing Values (Potential)	Numerical columns were set to use the median and categorical columns used the mode if missing values appeared.	Ensures the code would still work correctly if missing values were added to the dataset later.	Cell [9]
Outliers in Grades	Grades above 100 were capped at 100 using <code>.clip()</code>	Keeps grades within a realistic range and prevents extreme values from affecting the analysis unfairly.	Cell [13]
Data Integrity	Checked for duplicate rows (none were found).	Confirms that each student is counted only once and no records are repeated.	Cell [10]

5. Feature Engineering

To strengthen the analysis and capture patterns that are not visible from single columns alone, three new features were created.

New Feature	How It Was Created	Why It Is Useful
Normalization	Gaming, study, and sleep hours were normalized using the formula: $(\text{value} - \text{mean}) \div \text{standard deviation}$	Places all time-based variables on the same scale, allowing them to be fairly compared and combined without one variable dominating due to larger values.
Academic Risk Score	Calculated as: Normalized Gaming – (Normalized Study + Normalized Sleep)	Combines multiple lifestyle behaviors into a single score that represents overall academic risk due to imbalance between gaming, study, and sleep.
Gaming Intensity	Gaming hours were grouped into Non-gamer, Light, Moderate, and Heavy categories	Allows clear comparison between different levels of gaming behavior and supports subgroup analysis instead of relying only on raw hour values.
Correlation Function	A custom function was created to calculate correlations and label them as strong/weak and positive/negative.	Simplifies relationship analysis, ensures consistent interpretation, and converts numerical correlations into clear insights.

6. Analysis Methodology

The analysis was carried out in clear and structured steps using the following tools:

Pandas:

Used to load the dataset, check for missing or duplicate values, and group data (for example, comparing averages by stress level).

NumPy:

Used for mathematical calculations such as normalizing data using the mean and standard deviation and creating trend lines using linear regression.

Matplotlib:

Used to create visualizations such as histograms, bar charts, scatter plots, pie charts, and boxplots to help identify patterns, relationships, and distributions in the data.

7. Analysis and Visualizations

[This is the main body of the report. Organize it by research question, not by code order. For each subsection, include the analysis, the relevant plot or table, and a short interpretation.]

7.1 Research Question 1

Does an "Academic Risk Score" predict grades better than individual factors like study or sleep hours alone?

To answer this, a comparative correlation analysis was performed using three distinct scatter plots: Academic Risk Score vs. Grades, Gaming Hours vs. Grades, and Sleep Hours vs. Grades. NumPy's polyfit function was utilized to generate linear regression trend lines and a custom interpret_correlation function to quantify the strength of these relationships.

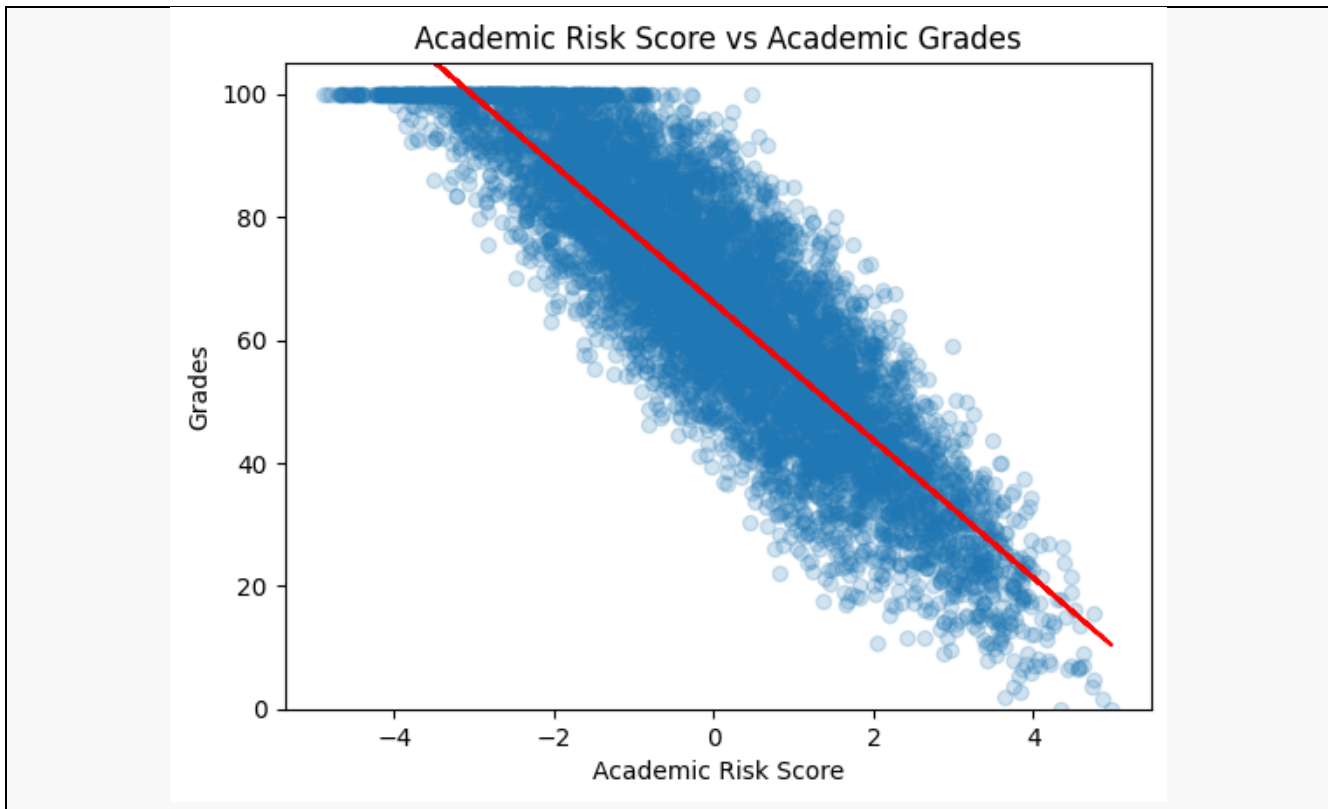


Figure 1. The scatter plot shows a clear negative relationship between academic risk score and academic grades. Students with higher risk scores tend to achieve lower grades, while students with lower or negative risk scores generally perform better academically. Although some variation exists, the overall trend supports the interpretation that excessive gaming combined with reduced study and sleep is associated with poorer academic outcomes.

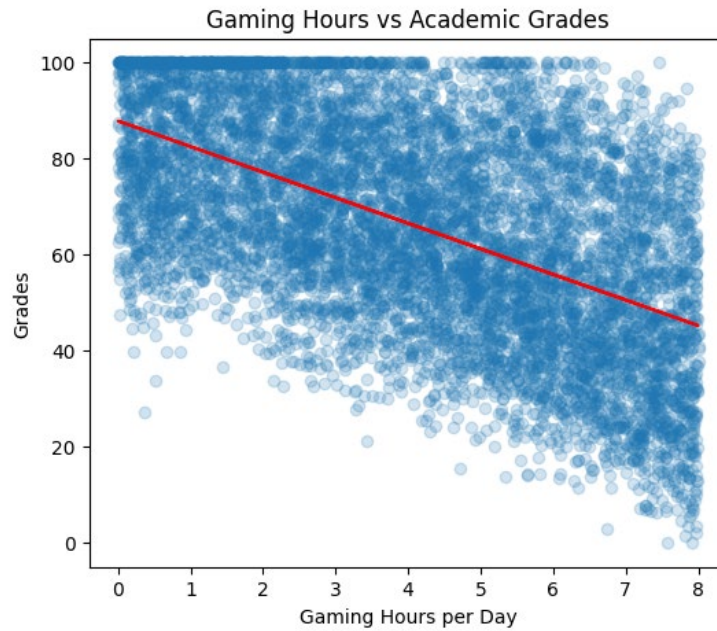


Figure 2. This scatter plot shows a weak negative correlation between gaming hours and academic grades, indicated by the downward trend line. Although the academic risk score was already analyzed, this plot is included to isolate and examine the direct relationship between gaming hours and grades without the influence of study and sleep.

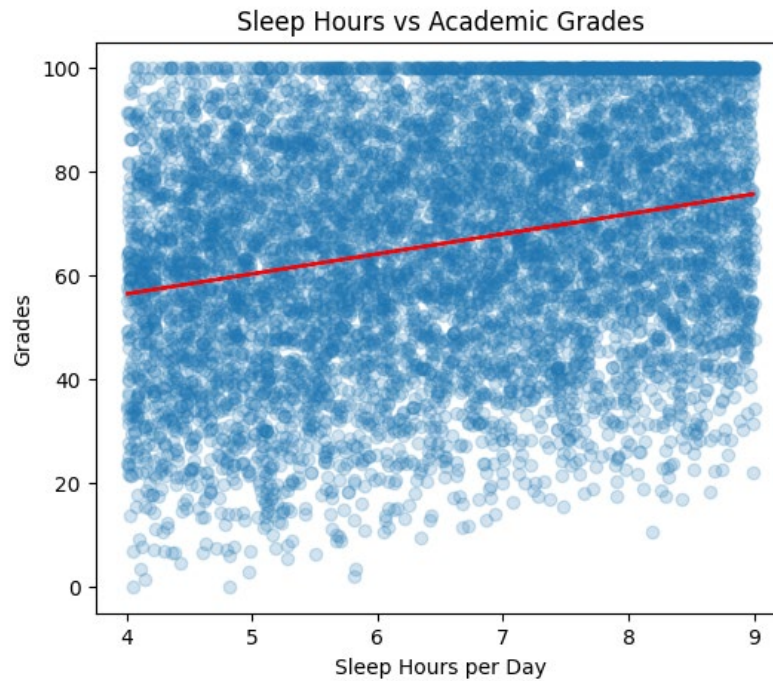


Figure 3. This scatter plot shows that students who sleep more tend to have higher grades on average, indicated by the upward trend line. The pattern suggests that sleep has an although positive yet weak relationship with academic performance.

Interpretation:

The results show that the Academic Risk Score is the strongest indicator of academic performance. It has a strong negative relationship with grades, meaning students with higher risk scores usually perform worse academically. When looking at individual factors, gaming hours alone have a moderate negative relationship with grades, while sleep hours show only a weak positive relationship.

This suggests that poor academic performance is rarely caused by one habit alone. Instead, it is more often the result of multiple habits combined, such as high gaming time together with low study and sleep. Overall, academic decline appears to be linked to lifestyle imbalance, rather than any single behavior on its own.

7.2 Research Question 2**Does improved reaction time from gaming translate into measurable benefits for academic performance?**

A two-step analysis was conducted to test this theory. First, Gaming Hours was correlated with Reaction Time to verify the cognitive benefit of gaming. Second, Reaction Time was correlated with Academic Grades to see if that benefit carried over into academics.

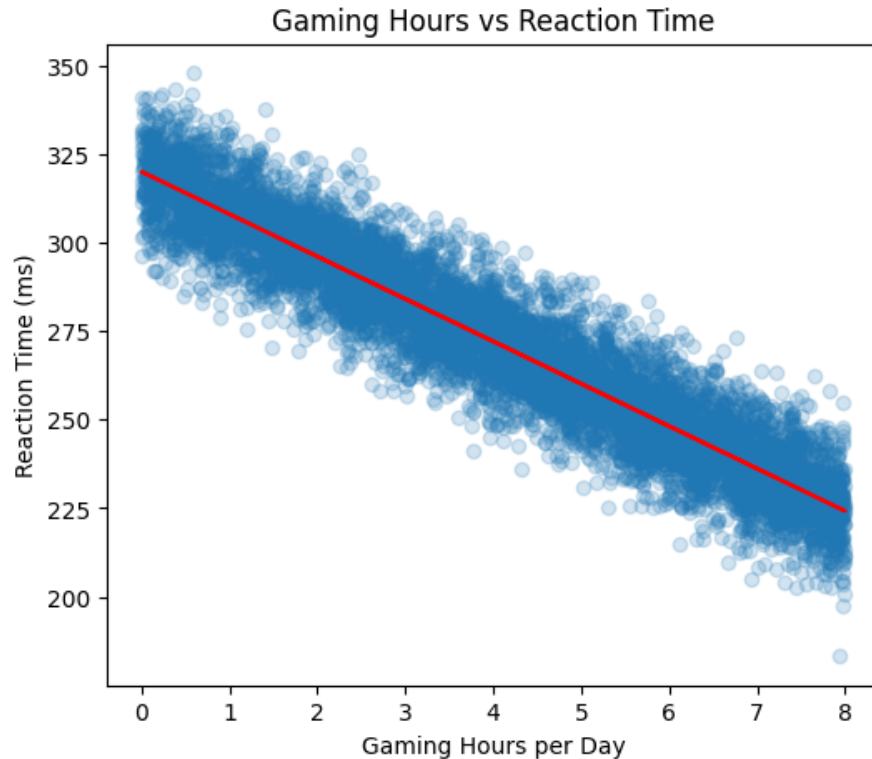


Figure 4. This scatter plot was created to check whether gaming helps improve cognitive skills, measured using reaction time. The strong negative correlation suggests that higher gaming hours are generally associated with faster reaction times, supporting the idea that gaming may improve certain cognitive abilities.

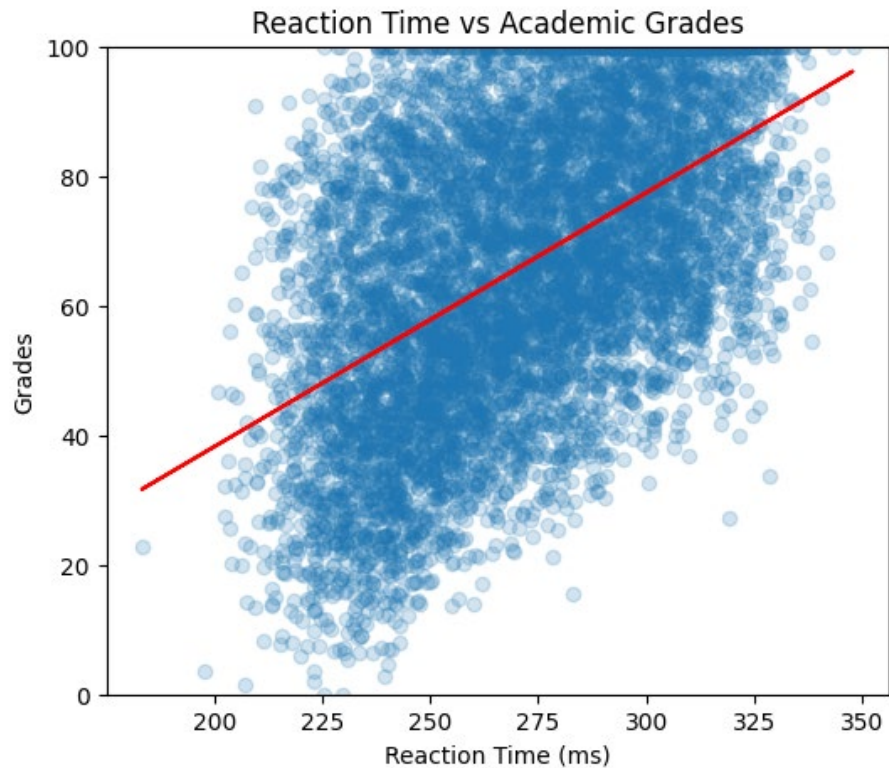


Figure 5. This scatter plot was created to see whether improved reaction time leads to better academic performance. Although the trend line slightly suggests that slower reaction times are linked to higher grades, the data points are widely scattered.

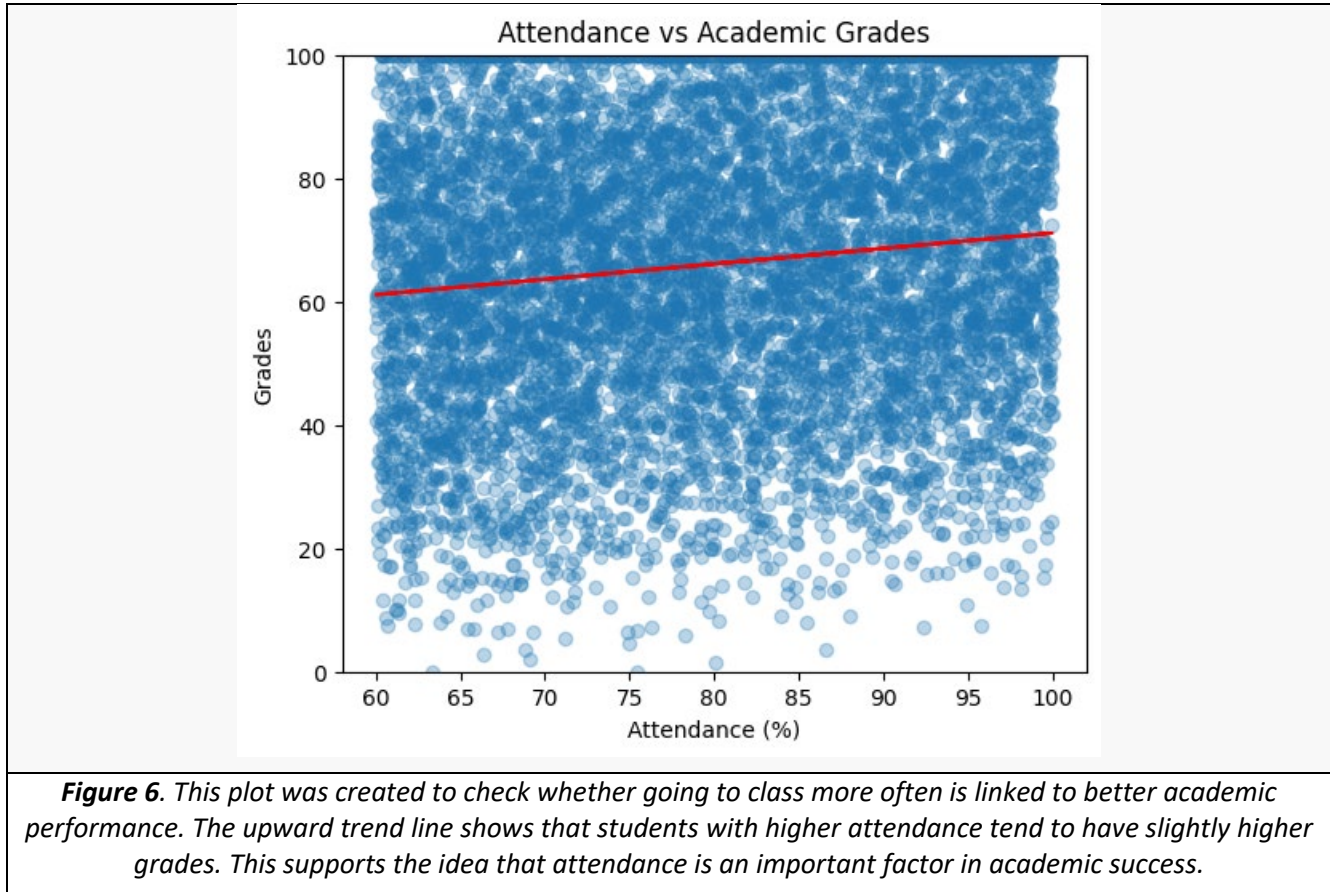
Interpretation:

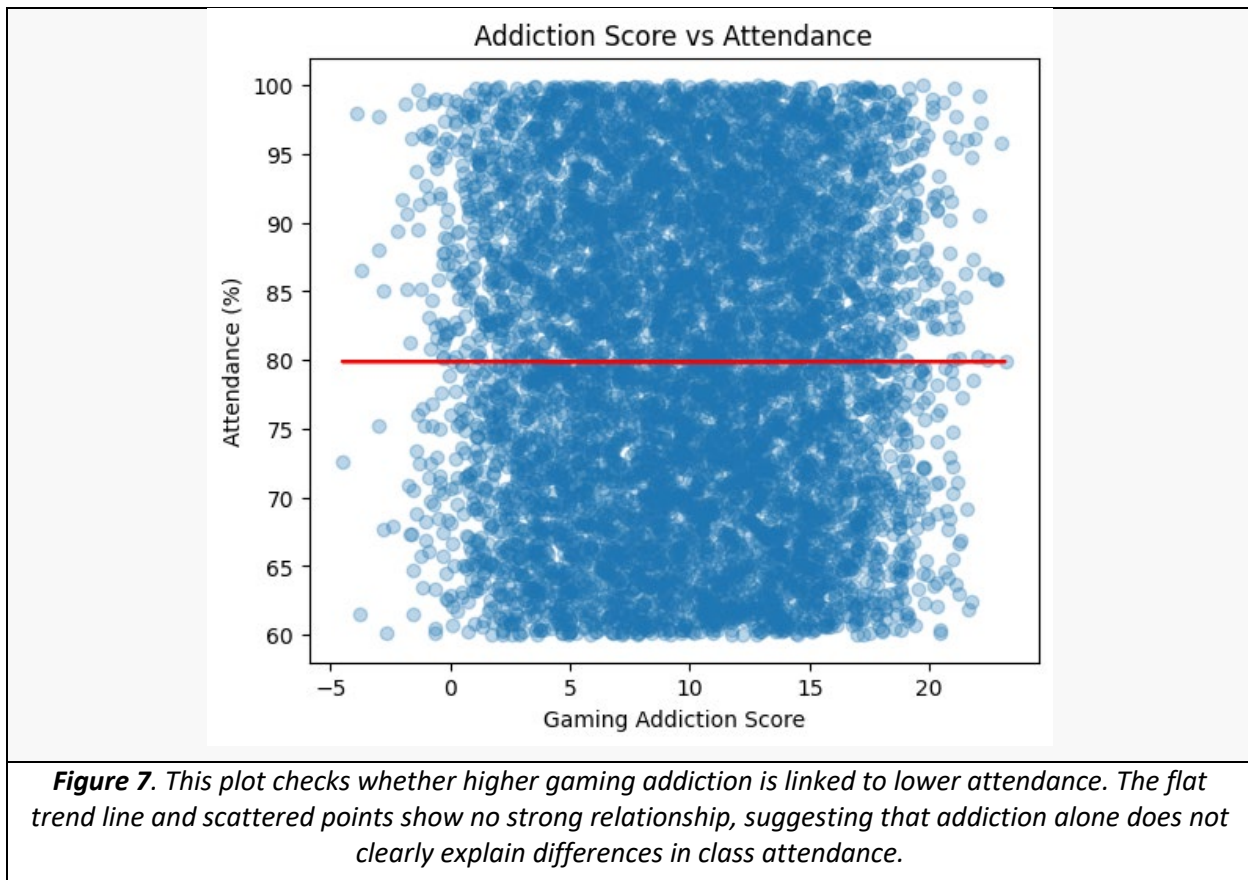
Gaming shows a clear cognitive benefit, as higher gaming is strongly linked to faster reaction time. However, this advantage does not lead to better academic results. The scatter plot between reaction time and grades is widely spread and shows only a weak relationship, suggesting that being faster at reacting in games does not translate into higher test scores or academic success.

7.3 Research Question 3

Does high gaming addiction correlate with a drop in school attendance?

This section investigates whether the negative academic impact of gaming is a result of in class. The analysis was conducted in two stages. First, verifying the relationship between attendance and grades, and second, checking if gaming addiction serves as a primary factor for absence.





Interpretation:

The analysis first established that attendance is a positive factor for success, as higher classroom presence aligns with slightly better grades. However, the investigation into Gaming Addiction vs Attendance revealed a correlation of ~ 0.0 , showing no measurable relationship between the two.

This suggests that gaming does not harm academic performance by causing students to skip class. Instead, the negative impact on grades appears to stem from lifestyle imbalances, where excessive gaming likely displaces critical hours for sleep and study.

7.4 Research Question 4

Does gaming reduce stress at the cost of grades, or does stress increase alongside poor academic performance?

To answer this, a comparative analysis was conducted by plotting average gaming hours by stress level and average academic grades by stress level. Pandas groupby operations were used to calculate mean values, allowing stress to be examined as a potential link between gaming behavior and academic performance.

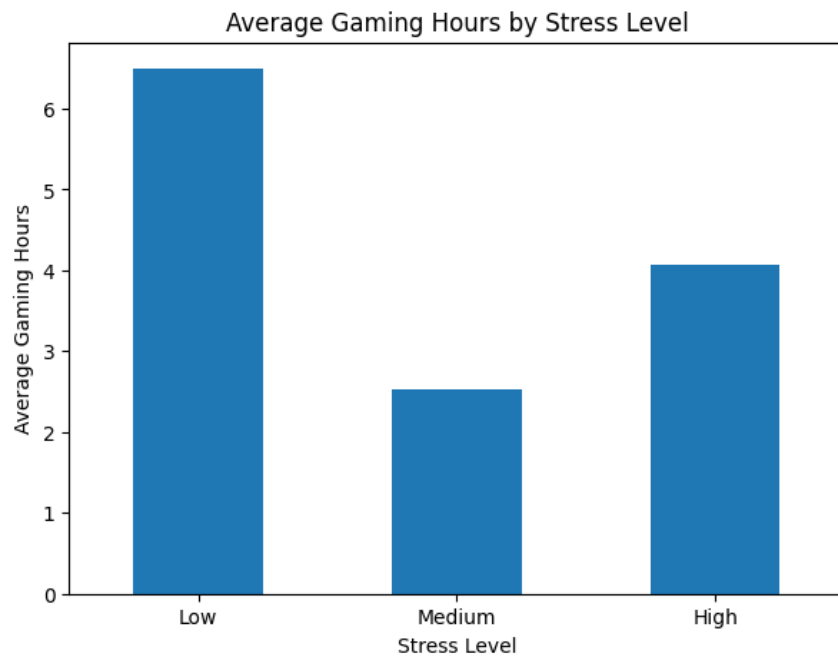
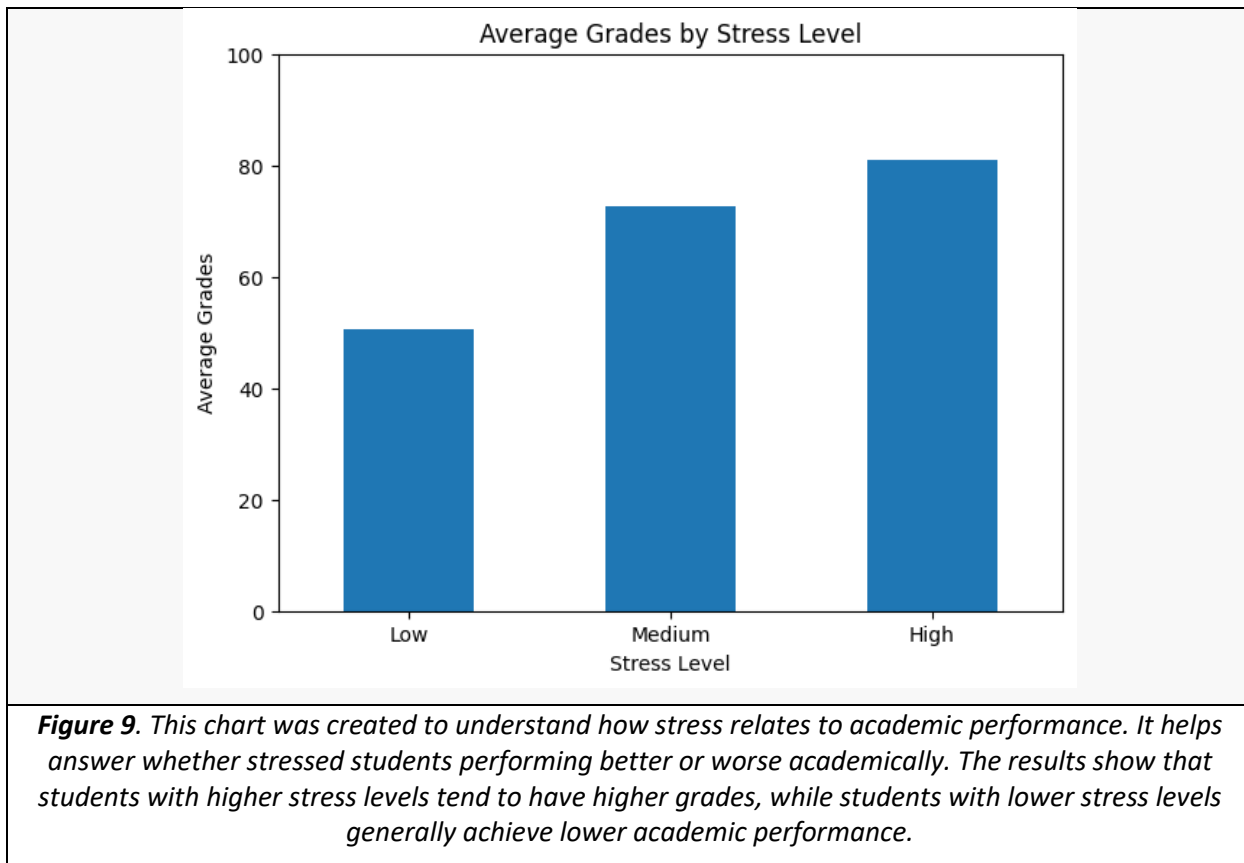


Figure 8. This chart was created to see how gaming behavior changes with stress. It helps answer whether more stressed students game more or less. The results suggest that students with lower stress levels tend to game more, while highly stressed students usually game in moderation. This indicates that gaming may help students relax, but excessive gaming is more common among those with lower stress.



Interpretation:

The results suggest that gaming and stress are related, but not directly. Students with lower stress levels tend to game more, which suggests that gaming may help some students relax. On the other hand, students with higher stress levels usually game in moderation and achieve higher grades. This indicates that academic success often comes with increased stress rather than reduced gaming.

Because higher gaming does not consistently lead to higher stress, stress does not appear to be a clear middle factor explaining why gaming is linked to poor grades. Instead, gaming, stress, and academic performance seem to interact independently, with academic performance being more strongly influenced by lifestyle balance and time management than by stress alone.

7.5 Research Question 5

Does the game genre a student plays affect academic performance?

A boxplot was used to visualize the distribution of grades across three major genres: Casual, FPS, and RPG. This allowed for a comparison of the median, quartiles, and range of grades for each category of gamer.

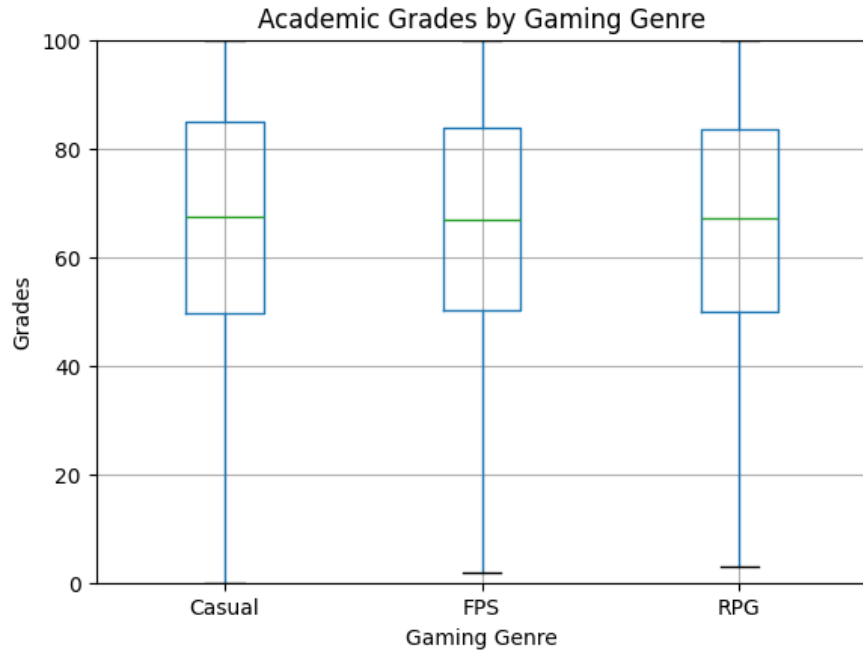


Figure 10. Boxplot illustrating the distribution of student grades categorized by the main gaming genre played.

Interpretation:

Academic performance appears largely consistent across different gaming genres, with no genre showing a clear advantage or disadvantage. This suggests that the type of game played is less important than how gaming fits into a student's overall routine. In other words, academic outcomes are shaped more by time management and lifestyle balance than by the specific content or genre of games.

8. NumPy-Based Custom Computation

[Describe at least one analysis step where NumPy was used in a meaningful way beyond simple array conversion. Examples include z-score computation, normalization, percentiles, np.where-based classification, binning, or a custom score.]

Component	Description
Computation used	Normalization and Academic Risk Score Calculation
Purpose	To standardize different lifestyle variables (gaming, study, and sleep hours) onto a common scale so they could be combined into a single predictive risk metric.
Formula or logic	The variables were normalized using the formula $Z = \frac{(x - \mu)}{\sigma}$ (where μ is the mean and σ is the standard deviation). The final score was calculated as: $\text{Academic Risk Score} = \text{Gaming}_{\text{norm}} - (\text{Study}_{\text{norm}} + \text{Sleep}_{\text{norm}}).$
Result / takeaway	Standardizing the data allowed multiple lifestyle factors to be combined into one score. This Academic Risk Score showed a much stronger negative relationship with grades (correlation of -0.88) than any individual factor on its own.

9. Key Findings

1. **Academic Risk Score Is Most Reliable:**

The Academic Risk Score was the strongest predictor of academic performance. Students with higher risk scores consistently achieved lower grades, making this combined measure more useful than any single factor alone.

2. **Cognitive Skill Does Not Equal Academic Success:**

Although increased gaming hours were strongly linked to faster reaction time, this cognitive improvement did not lead to better academic grades. Faster reactions in games do not directly improve academic outcomes.

3. **Attendance Is Not the Main Issue:**

Gaming addiction showed almost no relationship with class attendance. This suggests that students who game excessively are still attending classes, but may struggle academically due to poor lifestyle balance rather than absence.

4. **Stress and Performance Move Together:**

Students with higher stress levels tended to achieve higher grades, while lower-stress students usually gamed more and performed worse academically. This indicates that academic success often comes with ability to take increased stress.

5. **Gaming Genre Makes Little Difference:**

Academic performance was similar across all gaming genres. This shows that how much students game and how well they manage their time matters more than what type of games they play.

6. 10. Limitations

- **Self-Reported Data:**

Some information, such as gaming hours, study time, and sleep duration, is based on student reporting. This means the data may contain inaccuracies due to memory errors or students inaccurately recalling their own habits.

- **Correlation Does Not Mean Causation:**

Although strong relationships were found between certain variables (such as the Academic Risk Score and grades), this analysis cannot prove that gaming directly causes lower academic performance. It only shows that these factors are related.

- **Unmeasured Factors:**

The analysis does not include other influences such as family background, financial situation, learning difficulties, or the quality of study time, which could also affect academic performance.

- **Data Preprocessing Choices:**

Missing values were handled using median and mode, and grades were capped at 100. While these steps improved data consistency and realism, they may slightly reduce accuracy.

11. Conclusion

The project explored how different aspects of gaming relate to academic performance. The results show that while gaming can improve certain cognitive skills, such as reaction time, these improvements do not directly lead to better grades. Reaction time and academic performance showed a weak and scattered relationship, suggesting that cognitive speed alone is not an important factor in academic success.

The strongest finding of this analysis was the usefulness of the Academic Risk Score, which combined gaming hours, sleep, and study time. This combined score explained academic outcomes much more clearly than any single factor on its own. Students with a better balance of lower gaming and higher sleep and study time tended to perform better academically.

The analysis also showed that gaming addiction does not strongly affect class attendance, meaning that excessive gaming is unlikely to harm grades simply by keeping students out of class. Instead, the negative academic impact of gaming appears to be linked more to lifestyle imbalance, such as reduced sleep and study time, rather than the act of gaming itself.

Overall, the findings suggest that moderate gaming is not harmful, but maintaining balance is critical. Academic success in a gaming lifestyle depends on managing time, stress, sleep, and study habits rather than completely avoiding gaming.

12. References / Dataset Source

Dataset: Explore, A. (2024). Gaming vs Academic Performance. Kaggle.

<https://www.kaggle.com/datasets/aiexplorer77/gaming-vs-academic-performance>

Software: Developed using Python libraries: Pandas, NumPy, and Matplotlib

Final Submission Checklist

- ☒ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- ☒ I attached the project report as a separate file.
- ☒ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusion.
- ☒ My code uses Pandas, NumPy, and Matplotlib meaningfully.
- ☒ All bracketed placeholder text and helper notes were replaced or deleted before submission.